Supervised learning classification.

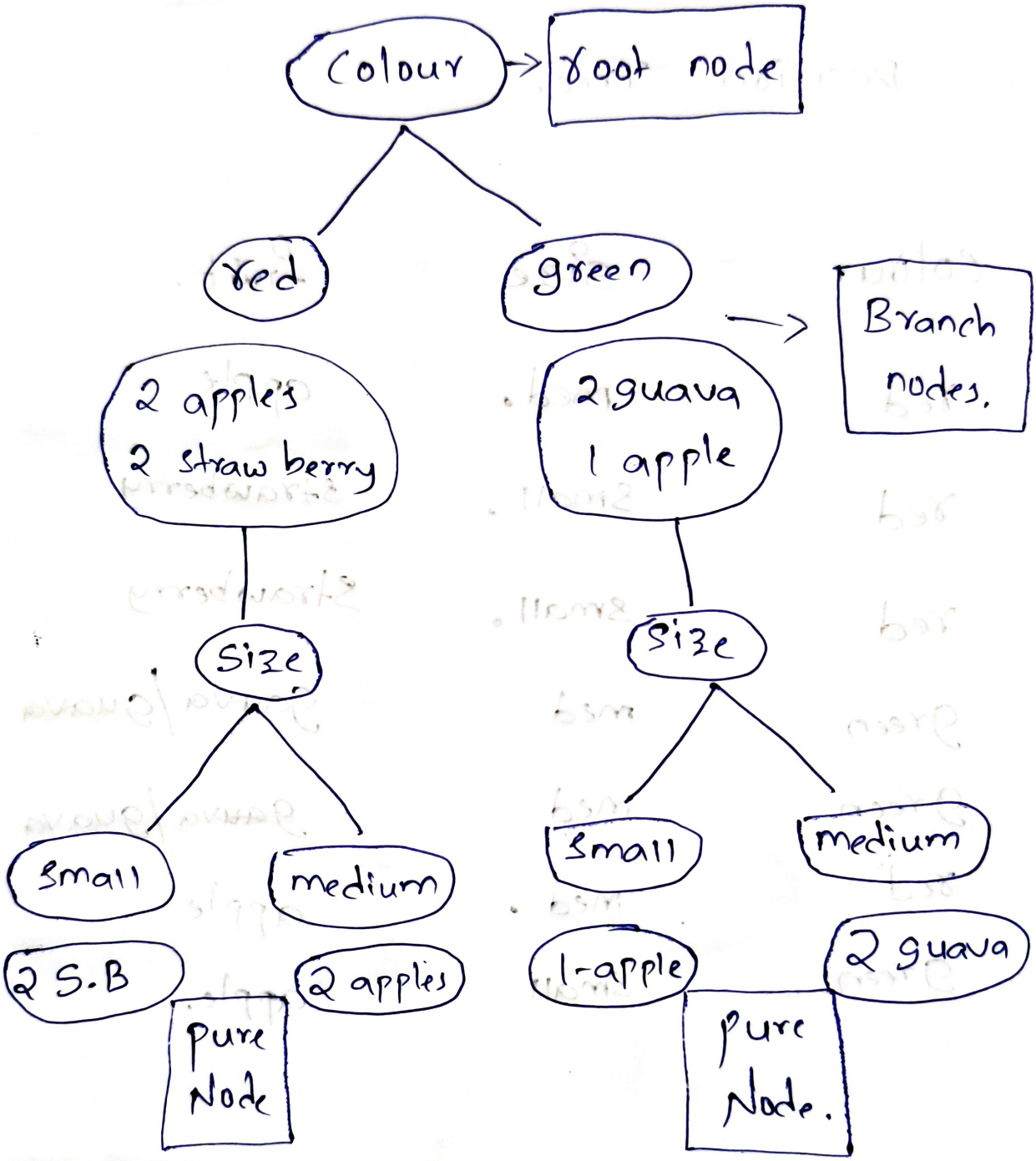
⊛. SLC.

→ Decision Tree.

Colour	Size	fruit.
red	med.	apple
red	small.	strawberry
red	small.	strawberry
green	med	~~gauva~~/guava
green	med	~~gauva~~/guava
red	med.	apple
green	small	apple.

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(3 apples, 2 S.B, 2 guava) 2 guava.



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* Basically Decision tree is a supervised learning algorithm that is used for classification and regression modelling.

* It is a tree based algorithm which consists of root node, branch node & pure Node.

* The important question here is how Decision tree is selecting the root node. for this we need to understand a concept called GINI & ENTROPY.

→ GINI & ENTRUPY both tell's me what is the Impurity / minture in the data.

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(*). Gini = $1 - \sum p_i^2$

(*) Entropy = $-\sum p_i \log_2 p_i$

(*) Now let's try to first understand how Gini works.

Past Trend (X_1)	Open Interest (X_2)	Trading volume (X_3)	Return (Target Var).
Positive	Low	High	up
Negative	high	low	down
positive	low	high	up
Positive	high	high	up
Negative	low	high	down
positive	low	low	down
Negative	high	high	down
Negative	low	high	down
positive	low	low	down
positive	high	high	up

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→ Now for the above example we need to calculate GINI Values for the Independent Variable w.r.t Targed Variable

→ Past trend vs Return.

Contingency table.

	up	down	
positive	4	2	⇒ 6
Negative	0	4	⇒ 4
	⇓	⇓	10
	4	6	

Gini (Past trend)

:- prob (positive) × Gini (positive) ⇒

+

prob (Negative) × Gini (Negative)

$prob(positive) = \frac{6}{10}$

$prob(Negative) = \frac{4}{10}$

Gini (positive) ⇒ $1 - \Sigma p_i^2 \Rightarrow 1 - \left[\left(\frac{4}{6}\right)^2 + \left(\frac{2}{6}\right)^2\right]$ 0.444

Gini (Negative) ⇒ $1 - \Sigma p_i^2 \Rightarrow 1 - \left[\left(\frac{0}{4}\right)^2 + \left(\frac{4}{4}\right)^2\right]$ 0

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$$\text{Gini (past trend)} = P(\text{pos}) \times G(\text{pos}) + P(\text{Neg}) \times G(\text{Neg})$$

$$= \frac{6}{10} \times 0.444 + \left(\frac{4}{10}\right) \times 0$$

$$= 0.2666$$

similarly the values for open interest and Trading volume are calculated.

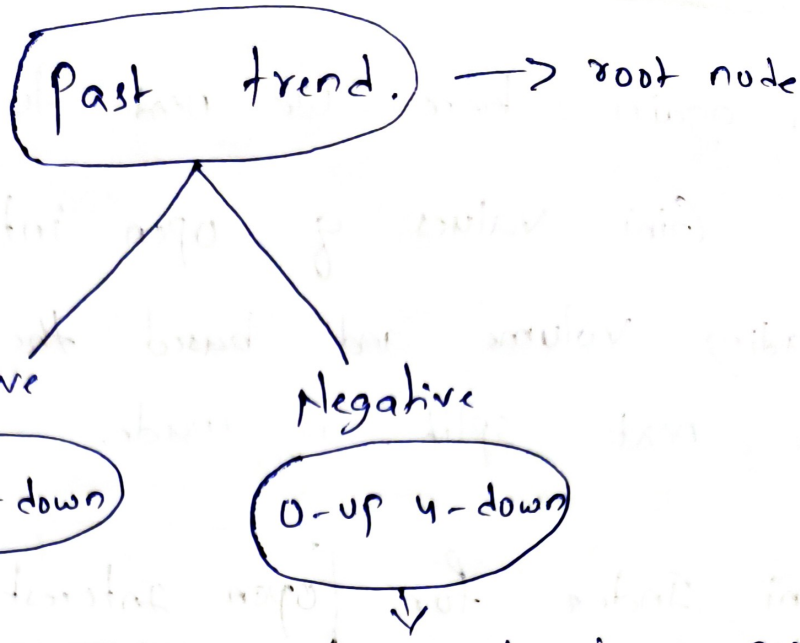
Gini (past trend) = 0.26

Gini (open interest) = 0.466

Gini (Trading Volume) = 0.342

Now here out of all the Variables the gini index of the past trend Variable is low so we will select past trend as root node and we will move forward.

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This branch is not a pure node so we need to further classify this so now we need to repeat all the steps again.

This node is a pure node because the node is completely classified & the Gini value becomes 0.

so the updated data will be

Past Trend	open Interest	Trading Volume	Return
positive	Low	High	up
positive	Low	High	up
positive	High	High	up
positive	Low	Low	down
positive	Low	Low	down
positive	High	High	up

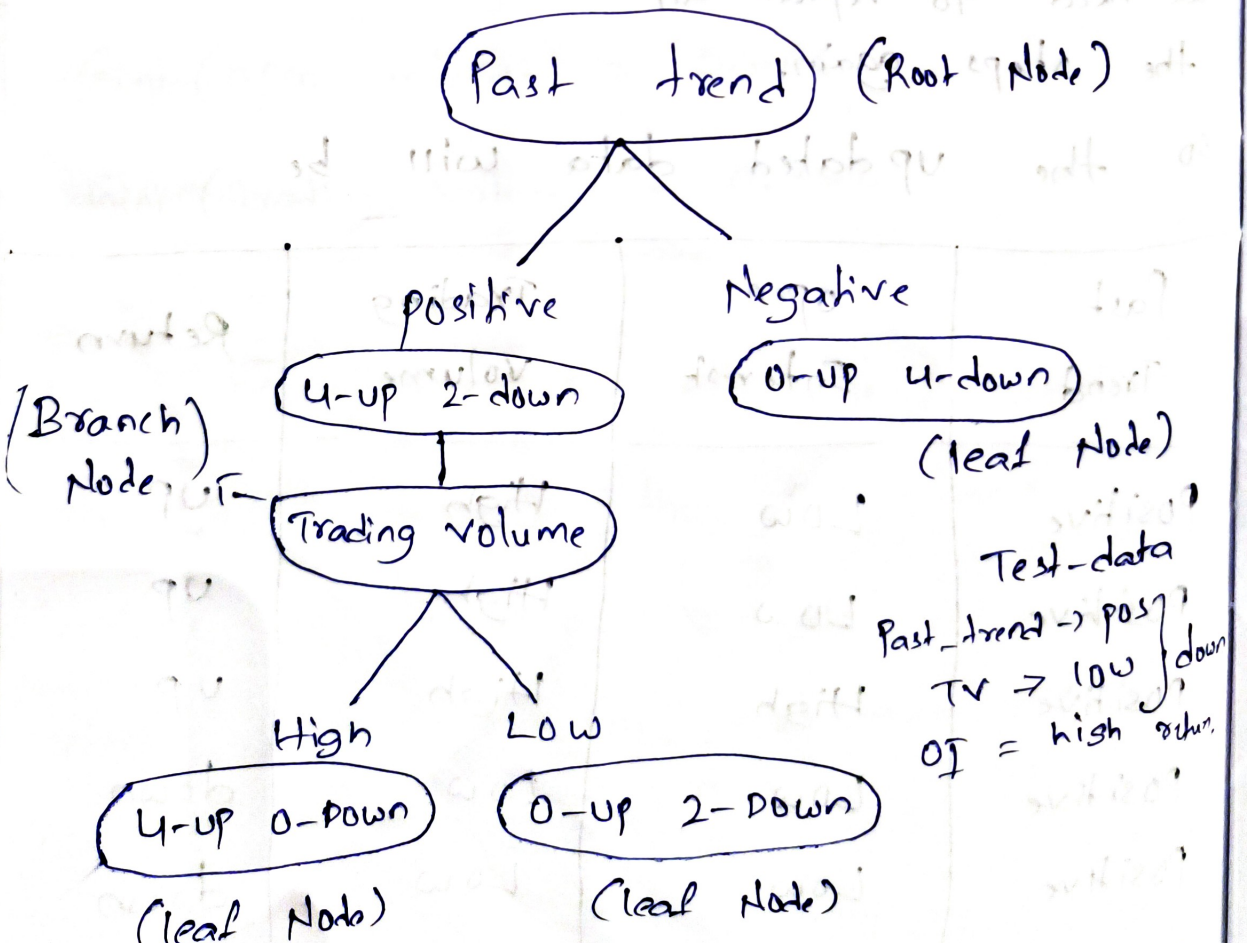
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Now again here we need to calculate the Gini values of open interest & Trading volume and based the Gini values the next split is made.

Gini Index for **open Interest = 0.33**

Gini for Trading **volume = 0**

Here, the split node will be Trading volume as its gini Index is low.



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Gini Gain = G(Y) - G(X/Y)

$G\left(\frac{\text{Past trend}}{\text{Return}}\right)$ = prob(pos) x Gini(pos)
+
prob(Neg) x Gini(Neg)

$G(\text{Return}) = 1 - \sum p_i^2 = 1 - \left[\left(\frac{4}{10}\right)^2 + \left(\frac{6}{10}\right)^2\right]$

$= 1 - (0.16 + 0.36)$

$= 1 - 0.52 = 0.48$

Gini Gain (past trend)

0.48 - 0.26 => 0.22

T.V	OI
0.48	0.48
0.34	0.46
0.14	0.2

Now here we can choose the root node basically by two ways either Gini Gain / Gini.

* Gini Gain :- The column with max Gini Gain is used as Root Node

* Gini :- The column with min Gini value is taken as Root Node.

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⊗ Now let's understand ENTROPY.

(X(Y)

⊗ Ent (Past trend / Return)

= prob (~~past trend~~) * ~~ent~~ (up

= prob (positive) × entropy (positive)

+

prob (Negative) × entropy (Negative)

	up	down	
P(+)	4	2	= 6
N(-)	0	4	= 4
			10

$$= \frac{6}{10} \times -\left[\frac{4}{6}\log_2\frac{4}{6} + \frac{2}{6}\log_2\frac{2}{6}\right]$$

+

$$\frac{4}{10} \times -\left[\frac{0}{4}\log_2\frac{0}{4} + \frac{4}{4}\log_2\frac{4}{4}\right]$$

$$= \frac{6}{10} \times 0.918 + \frac{4}{10} \times 0$$

$$= 0.5508.$$

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8. $Ent(Return)^{(y)}$

$$-\left[\frac{6}{10}\log_2\frac{6}{10} + \frac{4}{10}\log_2\frac{4}{10}\right]$$

$$-\left[0.6\log_2 0.6 + 0.4\log_2 0.4\right]$$

$$-0.6\log_2 0.6 - 0.4\log_2 0.4$$

$$-[-0.133 + (-0.15)] \Rightarrow -[-0.133 - 0.15]$$

$$= -0.6 \times (-0.737) - 0.4 \times (-1.322)$$

$$0.4422 + 0.5288$$

$$= 0.971..$$

Here also we need to remember that

Information Gain :- Entropy (y) − Entropy (x/y)

Entropy (Return) = 0.971

Entropy (Past trend/Return) = 0.55

$$IG = 0.971 - 0.55 \Rightarrow 0.41$$

similarly all the columns Entropy / IG. needs to be calculated.

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⊗ Now Important point if we allow the decision tree to grow completely than the DT model will end up in overfit zone.

→ So inorder to avoid the overfit of model we need to perform hyper-parameter tuning. called PRUNING.

Disadvantage:-

→ prone to overfit

→ Greedy Nature.

⇒

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⊛. Decision Tree Regressor / Random Forest Regressor.

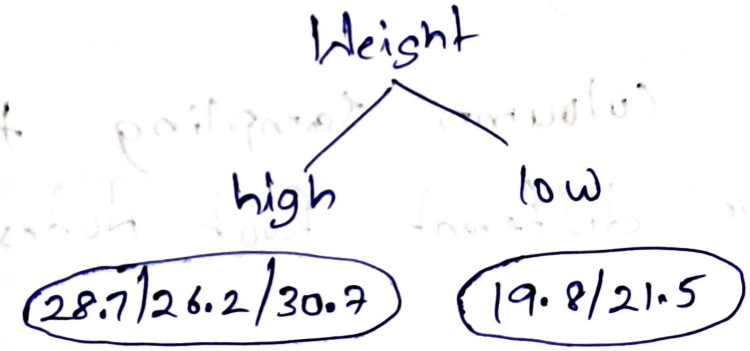
→ How do we calculate Root Nodes

↓↓

Gini, Entropy ⇒ Classification.

Root Node → MSE ⇒ Regressor.

Weight	bmi
high	28.7
high	26.2
low	19.8
low	21.5
high	30.7



MSE:- 2.5 MSE = 1.2

$$MSE = \sum (x - \bar{x})^2 / n$$

$$\text{Total MSE} = \frac{3}{5} \times (2.5) + \frac{2}{5} \times (1.2)$$

Total gini = $P(H)\,G(H) + P(L)\,G(L)$ (CAT)

⇒ $P(H)\,MSE(H) + P(L)\,MSE(L)$ (CON).

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